TD-PCL PROJECT REPORT

(Transdisciplinary Group Project)

EFFECTS OF CANCER ON PATIENTS' PSYCHOLOGICAL WELL-BEING

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- **Academic Level**: Postgraduate
- **Academic Year & Semester**: 2025 - 26, Semester III

Submitted to:

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Project Outcome: Integrated psychological, biological, and predictive-model insights into the mental-health burden experienced by cancer patients.

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1. Project Title: Effects of Cancer on Patients' Psychological Well-Being.
2. Theme (Transdisciplinary Area): Mental health of Cancer Patients
3. Academic Level: Postgraduate
4. Academic Year & Semester: 2025–2026, Semester III
5. Departments Involved:

- Computer Science
- Library & Information Science
- Psychology
- Media Studies
- Management

6. Mentor / Faculty Guide: Dr. Neha Priyadarshini

7. Executive Summary:

This interdisciplinary review explores how cancer affects patients' psychological well-being by integrating perspectives from psychology, biotechnology, and data science. Cancer represents not only a physiological illness but also a profound psychological journey marked by anxiety, depression, fear of recurrence, body-image changes, financial toxicity, identity disruptions, and social strain.

The project synthesizes evidence from more than twenty global studies to highlight:

- The multifaceted nature of psychological distress
- Its strong links to treatment burden, financial strain, and uncertainty
- The need for holistic, interdisciplinary intervention models

Biotechnology contributes insights on inflammatory pathways, biomarkers, fatigue mechanisms, and stress physiology. Psychology explains emotional responses, coping patterns, and therapeutic solutions such as mindfulness, cognitive-behavioural interventions, and positive psychology. Data science adds predictive analytics through PROMIS scores, AI-based monitoring, and ML models to identify high-risk patients.

Overall, the review demonstrates that improving psychological outcomes in cancer care requires integrated strategies across clinical, psychological, technological, and social domains.

8. Problem Statement & objectives:

Problem Statement

Cancer patients frequently experience severe psychological distress—manifesting as anxiety, depression, fear of cancer recurrence, body-image disturbance, and financial toxicity. These issues often remain under-recognized despite significantly affecting treatment adherence, recovery, and overall quality of life.

Objectives

- To systematically review psychological challenges encountered by cancer patients across demographics and cancer types.

- To integrate psychological, biotechnological, and data-science perspectives to propose holistic and evidence-based solutions for early detection and management of psychological distress.

9. Methodology & Approach

This project adopts a **systematic narrative review** framework, integrating three academic disciplines:

Psychology

- Review of validated assessment tools: HADS, CES-D, PROMIS, Distress Thermometer
- Analysis of interventions: ConquerFear, mindfulness, positive psychology, online counseling
- Study of emotional, cognitive, and behavioral mechanisms

Biotechnology

- Exploration of biological pathways linking stress with cancer progression
- Study of fatigue pathways, inflammatory markers, and treatment burden
- Interpretation of tumor microenvironment changes associated with psychological states

Data Science

- Cross-study comparison of statistical analyses, ML models, and predictive frameworks
- Extraction of distress predictors based on demographic, clinical, and psychosocial factors
- Review of AI tools using facial-expression analysis, PROMIS scoring, and risk-prediction models

This interdisciplinary method enabled a unified understanding of cancer-related psychological distress from clinical, cognitive, and computational standpoints.

10. Outcomes of the Project:

Outcome Type: Research Publication (In Preparation)

Summary

The project produced comprehensive interdisciplinary review synthesizing research from 2007–2025. Key outcomes include:

- Identification of major predictors of psychological distress
- Mapping gaps in screening and intervention delivery
- Highlighting effective psychological and digital interventions
- Proposing future research and predictive models for early triaging

The review is being prepared for submission to a peer-reviewed psycho-oncology or multidisciplinary health-science journal.

11. Innovation & Impact

Novelty

This project uniquely integrates **psychology, biotechnology, and data science** to examine cancer-related psychological distress. It progresses beyond symptom description and connects:

- Emotional & behavioral mechanisms
- Biological processes (inflammation, stress pathways)
- Machine-learning prediction models

Impact

- **For Patients:** Early detection and personalized support
- **For Clinicians:** Evidence-based screening and interdisciplinary intervention models
- **For Academia:** A multi-domain conceptual framework for future psycho-oncology studies
- **For Healthcare Systems:** Potential for AI-driven digital triaging and monitoring tools

12. Interdisciplinary Collaboration

Psychology

- Led conceptualization of emotional distress, coping, patient behavior
- Evaluated psychological tools and interventions

Biotechnology

- Analysed stress-related biological pathways, treatment impact, and tumor environment

Data Science

- Extracted patterns from reviewed studies
- Evaluated predictive analytics, AI models, and statistical methodologies

Together, these disciplines created a cohesive, multi-layered review.

13. Challenges Faced & How They Were Overcome

- Wide variability in study designs: Managed by grouping studies based on themes and outcome measures.
- Different disciplinary languages (biotech, psychology, data science): Resolved by creating common definitions and integrating terminology.
- Limited Indian-context studies: Addressed by carefully highlighting available Indian research and comparing it with global evidence.
- Overlapping psychological constructs (anxiety, depression, FCR): Clarified by using standardized definitions and validated scales

14. Future Scope / Roadmap:

- Develop a **predictive distress-risk model** using machine-learning based on factors identified in the review.
- Design an **online psycho-oncology intervention module** integrating counseling, symptom tracking, and AI-based alert systems.
- Conduct multicenter primary research on Indian populations.
  - Explore biological markers of distress (e.g., cortisol, inflammatory cytokines) for personalized care.
  - Publish the review to guide future interdisciplinary collaborations in cancer care.

15. Declaration by the Team

We, the undersigned, hereby declare that the project work titled "EFFECTS OF CANCER ON PATIENTS' PSYCHOLOGICAL WELL-BEING" is original and carried out by us under the guidance of faculty, following ethical research practices.

Attachments

- Data collection [research papers and their summary]

Title	Brief Description	Reference
The impact of fear of cancer recurrence on anxiety and depression in women with cancer: The chain mediating roles of financial toxicity and psychological distress — A multicenter investigative study	This multicenter investigative study analyzed female cancer patients to explore the relationship between fear of cancer recurrence (FCR), financial toxicity (COST), psychological distress (DT), anxiety, and depression (HADS). It identified an indirect chain mediation pathway where FCR exacerbates financial toxicity, which then intensifies psychological distress, ultimately contributing to anxiety and depression.	Liu, L., Liu, M., Liu, Y., Yi, H., Lyu, Z., Xing, S., & Liu, Y. (2025). *Asia-Pacific Journal of Oncology Nursing, 12,* 100699. doi:10.1016/j.apjon.2025.100699.
Psychological intervention improves quality of life in patients with early-stage cancer: a systematic review and meta-analysis of randomized clinical trials	This systematic review and meta-analysis investigated the effect of psychological interventions (PI) on survival (OS/RFS) and quality of life (QoL) in cancer patients. The authors concluded that PIs do not prolong survival (OS: HR = 0.97; RFS: HR = 0.99) but significantly improve QoL across global (SMD = 0.65), emotional (SMD = 0.64), social (SMD = 0.32), and physical (SMD = 0.33) domains, particularly in early-stage and breast cancer patients.	Bognár, S. A., Teutsch, B., Bunduc, S., Veres, D. S., Szabó, B., Fogarasi, B., Zahariev, O. J., Vörhendi, N., Almog, O., Hadani, Y., Gergő, D., Mihály, E., Erőss, B., Bunduc, S., Márta, K., & Hegyi, P. (2024). *Scientific Reports.*
Mental Health Challenges in Cancer Patients: A Cross-Sectional Analysis of Depression and Anxiety	This cross-sectional study evaluated the prevalence of depression and anxiety symptoms (using PROMIS self-report scales) in cancer patients and aimed to identify associated risk factors. Key risk factors identified for mental health issues included metastatic cancer, being unmarried/single, and having a past mental health disorder.	Shalata, W., Gothelf, I., Bernstine, T., Michlin, R., Tourkey, L., Shalata, S., & Yakobson, A. (2024). *Cancers, 16,* 2827. doi:10.3390/cancers16162827.
Research progress on the psychological burden and intervention measures in cancer patients	This review summarized the psychological burdens (e.g., anxiety, depression, demoralization, PTSD) and high-risk factors worldwide, outlined traditional assessment methods (CES-D, PHQ-9, HADS), and presented new developments like AI predictive models based on facial expressions or neuroimaging. It advocated for multidisciplinary and multidimensional comprehensive management.	Wang, H-Q., Lin, H., & Liu, B. (2024). *Frontiers in Psychiatry, 15,* 1408762. doi:10.3389/fpsyt.2024.1408762.

Managing Body Image Difficulties of Adult Cancer Patients: Lessons from Available Research	A review identifying empirically-supported approaches to treat body image difficulties in adult cancer patients. It synthesized literature (2003–2013) on appearance alterations, sensory changes, and functional impairment, and suggested a practical framework for assessment and treatment, including conversation strategies for the oncologic healthcare team.	Fingeret, M. C., Teo, I., & Epner, D. E. (2014). *Cancer, 120*(5), 633–641. doi:10.1002/cncr.28469.
Cancer-Related Fatigue and Its Associations with Depression and Anxiety: A Systematic Review	A systematic review assessing 59 studies (1950–2007) to quantify the associations between Cancer-Related Fatigue (CRF), depression, and anxiety. It concluded that depression is consistently and moderately associated with CRF (weighted average R = 0.56), while anxiety is also correlated, though at a somewhat lower magnitude (weighted average R = 0.46).	Brown, L. F., & Kroenke, K. (2009). *Psychosomatics, 50*(5), 440–447. doi:10.1176/appi.psy.50.5.440.
Financial toxicity and psychological distress in adults with cancer: A treatment-based analysis	A multicenter cross-sectional study in China to evaluate disparities in financial toxicity (FT) and psychological distress (PD) across different treatment groups (chemotherapy, adjuvant therapy, and surgery). It found high rates of FT (47.9%) and PD (56.5%), noting that surgery/adjuvant therapy correlated significantly with high PD scores.	Yu, H., Li, H., Zuo, T., Cao, L., Bi, X., Xing, H., Cai, L., Sun, J., & Liu, Y. (2022). *Asia-Pacific Journal of Oncology Nursing, 9*(9), 100069. doi:10.1016/j.apjon.2022.04.008.
Spirituality and its association with psychological distress in cancer patients attending a tertiary care hospital in Bangalore, India	A cross-sectional study examining the association between spirituality and psychological distress (anxiety and depression) in cancer patients in India. It observed a negative correlation between spirituality and depression (r = −0.833) and anxiety (r = −0.631), suggesting spirituality may help control psychological distress.	Vidya, D. C., Gangekar, S., Shalini, C., Harish, K., & Murthy, N. S. (2016). *Int J Community Med Public Health, 3*(7), 1812–1815. doi:10.18203/2394-6040.ijcmph20162047.
Assessment of psychological distress and its association with socio-demographic variables among cancer patients in a tertiary care hospital in Bengaluru, Karnataka: A cross sectional study	A cross-sectional study using the Distress Thermometer (DT) to measure psychological distress among cancer patients. It found that 32–43% experienced moderate to severe distress, with emotional concerns (anxiety, depression, worry, fear) being the most prevalent problems.	Pooja, S., Savatagi, S., Sengupta, A., & Kurunamurthy, J. (2023). *Int J Community Med Public Health, 10*(9), 3358–3366. doi:10.18203/2394-6040.ijcmph20232704.

Exploring the problems faced by patients living with advanced cancer in Bengaluru, India	An explorative study using thematic analysis based on semi-structured interviews with 10 advanced cancer patients in Bengaluru, India. Prevalent themes identified were financial instability, hopelessness, family anguish, self-blame, anger, stress, and suicidal thoughts, stressing the need for a holistic approach focused on the patient.	Wajid, M., Rajkumar, E., Romate, J., George, A. J., & Lakshmi, R. (2021). *Heliyon, 7*(4), e06686. doi:10.1016/j.heliyon.2021.e06686.
Exploring the Role of Online Psycho-Oncology Counseling Interventions in Alleviating Distress Among Cancer Patients	A study evaluating the effectiveness, feasibility, and satisfaction of Online Psycho-Oncology Counseling (OPC) interventions for cancer patients. The OPC program resulted in a significant reduction in mean distress scores (from 5.67 to 4.77).	Jaitri Mandal, R. K., Mukherjee, A., Govindraj, K., Veldore, S. V. H., Kshitiij D. R., & Goswami, H. M. (2024). *Asian Pac J Cancer Care, 9*(3), 453–461. doi:10.31055/APJCC.2024.9.3.453.
The Impact of Breast Cancer on the Patient and the Family in Indian Perspective	An observational mixed-methods study (9 years) exploring the financial burden, family involvement, decision-making roles (often male-led), and psychosocial issues (e.g., social embarrassment due to hair loss) of women with breast cancer in India.	Alexander, A., Kaluve, R., Prabhu, J. S., Korlimarla, A., Srinath, B. S., Manjunath, S., et al. (2019). *Indian J Palliat Care, 25,* 66–72. doi:10.4103/IJPC.IJPC_158_18.
'Patients with Cancer are Distressed!' Indian Healthcare Provider Perspectives on Distress Screening and Referrals to Psycho-oncology Services – A Mixed Methods Study	A mixed-methods study investigating distress screening practices in a tertiary cancer center. It found that 53.5% of new patients had moderate to extreme distress (DT ≥ 4), but only 8.6% were referred to Psycho-Oncology Services (POS), highlighting significant underutilization despite provider awareness of distress.	Normen, M., Sahaya, F. E., Kulkarni, K., Vidhubala, E., Shewade, H. D., & Kathiresan, J. (2021). *Indian J Palliat Care, 27*(4), 561–570. doi:10.25259/IJPC_41_21.
Positive psychology interventions in breast cancer: A systematic review	A systematic review that synthesized evidence on positive-psychology interventions (e.g., mindfulness, spiritual interventions) for women with breast cancer. It showed promise for improving psychological well-being and meaning-related outcomes but emphasized the need for larger, rigorously designed RCTs.	Casellas-Grau, A., Font, A., & Vives, J. (2014). *Psycho-Oncology, 23*(1), 9–19. doi:10.1002/pon.3353.

Introducing psychology services to advanced cancer patients: A randomized double-blind trial	An RCT investigating the effect of introducing visible psychology services via video vignettes to advanced cancer patients. Exposure to the vignette including psychology services improved patients' willingness to accept and refer to psychological care.	Ann-Yi, S., et al. (2019). *Psycho-Oncology, 28*(10), 1978–1986. doi:10.1002/pon.5177.
Humanistic health management and cancer: Associations of psychology, nutrition, and exercise with cancer progression and pathogenesis	A narrative/scoping review arguing that psychological stress and maladaptive coping interact with biological processes (e.g., inflammation) to influence tumor environment and outcomes. It advocates for a humanistic health management model integrating psychology, nutrition, and exercise into cancer care.	Li, C., Zhang, J., Pan, P., Junjie, et al. (2024). *Advanced Science, 11*(22), e2400665. doi:10.1002/advs.202400665.
Describing and predicting psychological distress after colorectal cancer	A prospective cohort study that characterized the prevalence, course, and predictors of psychological distress following colorectal cancer diagnosis. It found that distress trajectories were influenced by younger age, comorbidities, poorer social support, and greater disease/treatment burden.	Lynch, B. M., Steginga, S. K., Hawkes, A. L., Pakenham, K. I., & Dunn, J. (2008). *Cancer, 112*(6), 1363–1370. doi:10.1002/cncr.23300.
ConquerFear-Group: A randomized controlled trial of an online-delivered group-based psychological intervention for fear of cancer recurrence in breast cancer survivors	An RCT evaluating an online, group-delivered adaptation of the ConquerFear intervention for breast cancer survivors with clinically significant FCR. The intervention produced clinically meaningful reductions in FCR and improved coping strategies.	Tauber, N. M., O'Toole, M. S., Jensen, A. B., Butow, P. N., et al. (2023). *Psycho-Oncology, 32*(4). doi:10.1002/pon.6193.
Psychological distress and associated additional medical expenditures in adolescent and young adult cancer survivors	A cross-sectional analysis quantifying the association between psychological distress and healthcare utilization among Adolescent and Young Adult (AYA) cancer survivors. It concluded that distressed AYAs incurred substantially higher annual medical expenditures and more office visits/prescriptions than non-distressed peers.	Abdelhadi, O. A., Pollock, B. H., Joseph, J. G., & Keegan, T. H. M. (2022). *Cancer, 128*(7). doi:10.1002/cncr.34064.
The impact of a personal cancer diagnosis on the psychological health of adolescent/young adult cancer survivors: A mixed methods study	A mixed-methods study investigating the psychological health of Adolescent/Young Adult (AYA) cancer survivors. Findings identified persistent psychological health needs across the cancer trajectory and emphasized the importance of age-appropriate coping and social connectedness.	Rao, P., Devine, K. A., Bingen, K., & Scott, A. M., et al. (2025). *Psycho-Oncology, 34*(9), e70272. doi:10.1002/pon.70272.

Cancer risk elicitation and communication: Lessons from the psychology of risk perception	A conceptual and evidence-review paper that examined methodologies of cancer risk elicitation and communication. It found that risk perceptions are heavily influenced by message framing, numeracy, and emotional reactions, leading to potential miscommunication and poor health decisions.	Klein, W. M. P., & Stefanek, M. E. (2007). *CA: A Cancer Journal for Clinicians, 57*(3), 147–167. doi:10.3322/canjclin.57.3.147.
Psychological distress, understanding of cancer and illness uncertainty in patients with Cancer of Unknown Primary	A mixed-methods study that examined psychological distress, illness understanding, and illness uncertainty in patients with Cancer of Unknown Primary (CUP). The study revealed high levels of psychological distress and significant gaps in understanding, where illness uncertainty correlated with greater distress.	Wolyniec, K., et al. (2022). *Psycho-Oncology, 31*(11), 1869–1876. doi:10.1002/pon.5990.

Brief Summary

This project focuses on developing a comprehensive research-sharing platform designed to streamline how university students discover, contribute to, and engage with academic work. The objective is to bridge the gap between student researchers and accessible academic resources by creating a centralized, user-friendly digital space. The relevance of this project lies in addressing common challenges such as limited visibility of student research, difficulty accessing domain-specific studies, and the absence of collaborative knowledge-building tools within university environments.

The project integrates multiple disciplines—including psychology, computer science, design, data analysis, and user experience—allowing each team member to contribute domain-specific insights. This interdisciplinary approach ensured that the platform was conceptually strong, technically feasible, visually intuitive, and grounded in an understanding of student needs and research behaviors.

The impact of the project is significant: it enhances academic participation, encourages peer-to-peer learning, supports research skill development, and promotes a culture of knowledge sharing within the university community. By creating a structured, accessible platform, the project lays the groundwork for long-term innovation in how students interact with research and collaborate academically.

Drive Link: https://drive.google.com/drive/folders/1bSRIRZgrNS8HKD72t1O4zooD84R23KDl